

A Mamdani Fuzzy Logic Controller for Environmental Regulation in a Smart Assistive Apartment: Design, GA Optimisation, and Benchmark Evaluation

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Abstract

Automating environmental control in assisted-living spaces demands a controller that can reason under uncertainty with heterogeneous sensor inputs. This report presents a Mamdani Fuzzy Logic Controller (FLC) built for a smart apartment occupied by a disabled resident. Six crisp sensor readings, covering temperature, humidity, external luminance, time of day, occupancy count, and physical activity level, are fuzzified and processed through a 25-rule inference engine to drive three actuators: HVAC, dimmable lighting, and motorised blinds. A real-coded Genetic Algorithm (GA) subsequently optimises the 76-parameter membership function (MF) encoding and reduces system RMSE by refining MF breakpoints over 80 generations. A separate benchmark study then compares GA against Particle Swarm Optimisation (PSO) on two CEC 2005 functions across $D = 2$ and $D = 10$ dimensions, clarifying when each meta-heuristic is preferable. Results confirm that the FLC responds sensibly across a simulated 24-hour scenario and that GA-PSO relative performance is strongly landscape-dependent.

1. Introduction

Ambient Assisted Living (AAL) research has gained traction as ageing populations and rising disability prevalence create demand for homes that adapt to resident needs without requiring constant manual intervention [1]. Rule-based crisp systems cannot handle the linguistic vagueness inherent in descriptions like “too warm” or “adequate lighting”. Fuzzy Logic Controllers address this by mapping physical measurements to overlapping linguistic categories and combining rules through a principled inference mechanism.

The Mamdani inference model is chosen here because its output membership functions remain interpretable as linguistic descriptors such as “mild cooling” or “medium brightness”. This interpretability matters in a care context where occupational therapists, not engineers, may need to audit and modify rules. Sugeno models, while computationally leaner, map outputs to numeric functions that are harder to explain without domain expertise.

Three interlocking questions guide the work. First, can a relatively compact 25-rule Mamdani FLC produce contextually appropriate HVAC, lighting, and blind positions across the full daily cycle of a resident’s activity? Second, does a GA operating on MF breakpoints measurably improve FLC accuracy against a noisy reference dataset, and what does the chromosome structure look like? Third, how do GA and PSO compare on the well-studied CEC 2005 benchmark functions, and what does that comparison reveal about their inductive biases?

2. Part 1: FLC Design and Implementation

2.1. Problem Framing and Input-Output Selection

The target environment is a single apartment occupied by one mobility-impaired resident. Comfort depends on thermal conditions, light quality, and privacy from external glare. Monitoring uses six sensors listed in Table 2, selected because they jointly determine comfort without requiring complex wearable hardware. Three actuators are controlled: the HVAC unit, a dimmable light circuit, and motorised window blinds.

Table 1: FLC inputs and their crisp operating ranges.

Input	Unit	Range	Fuzzy Sets
Temperature	°C	0–40	cold / comfortable / hot
Humidity	%	0–100	dry / comfortable / humid
External light	lux	0–1000	dark / moderate / bright
Time of day	hour	0–24	night / morning / afternoon / evening
Occupancy	persons	0–5	empty / low / high
User activity	level	0–10	resting / moderate / active

Table 2: FLC inputs and their crisp operating ranges.

Output	Range	Fuzzy Sets
HVAC power	–100 to +100	strong_cool / mild_cool / neutral / mild_heat / strong_heat
Lighting level	0–100%	off / dim / medium / bright
Blinds position	0–100%	closed / half_open / open

2.2. Fuzzy Inference Model Selection

A **Mamdani** model is adopted. The core reason is human intelligibility. Because outputs remain linguistic fuzzy sets, a care worker can read a rule such as “*IF temperature is hot AND user_activity is active THEN hvac_output is strong_cool*” and immediately verify whether the behaviour matches professional guidelines. Sugeno models produce crisp polynomial outputs per rule, which are compact but lose the linguistic grounding that AAL stakeholders expect.

The inference chain follows four steps: (1) fuzzification maps crisp sensor values to membership degrees, (2) rule evaluation applies the **min** operator for AND-connected antecedents, (3) implication clips each consequent MF at the firing strength, (4) aggregation combines all clipped MFs with **max** before (5) centroid defuzzification converts the aggregate fuzzy set to a single crisp control value.

2.3. Membership Function Design

2.3.1. Input MFs

Trapezoidal MFs (**trapmf**) model the boundary sets of each input, providing flat plateaux where membership remains at 1.0 even as the physical measurement fluctuates. This is deliberate: a temperature of 5°C is unambiguously cold regardless of small noise, so the plateau prevents unnecessary HVAC toggling. Triangular MFs (**trimf**) model interior linguistic sets where gradual transitions between categories carry useful information.

The temperature input carries three MFs with parameters tuned to align with clinical thermal comfort standards. *Cold* spans 0–17°C (plateau 0–12°C), *comfortable* peaks at 19°C with support from 14°C to 24°C, and *hot* rises from 22°C upward with a plateau from 27°C to 40°C. Overlap between adjacent MFs is deliberately kept at roughly 30% of the triangle base so rules fire smoothly without abrupt switching artefacts.

Humidity follows the same three-class structure. External light is discretised into three classes covering the full 0–1000 lux range, with *moderate* peaking at 400 lux, consistent with indoor illuminance guidelines for occupied workspaces. Time of day carries four MFs because a residential day naturally divides into four behavioural phases: night-rest, morning routine, afternoon activity, and evening relaxation.

Occupancy and activity levels use compact ranges (0–5 persons and 0–10 respectively), so MF widths are narrower but the same overlapping trapezoid or triangle strategy applies.

Figure 1 shows all six input MF plots.

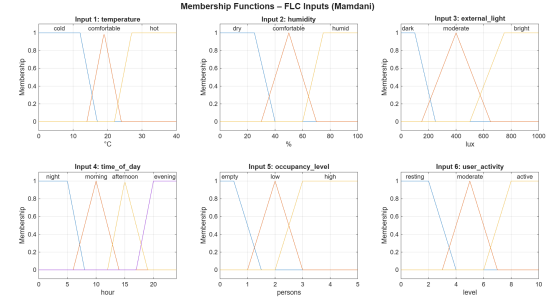


Figure 1: Input membership functions for all six sensor variables. Trapezoidal MFs model boundary linguistic labels; triangular MFs model interior transitions.

2.3.2. Output MFs

HVAC output uses five MFs to cover a bipolar range (−100 to +100). Negative values command cooling, positive values command heating. The *neutral* set is a narrow triangle centred at zero (support −10 to +10) so the actuator is instructed to remain off when conditions are already comfortable. Lighting uses four levels: *off*, *dim*, *medium*, and *bright*. Blinds use three positions. All output MFs use the same trapezoid-at-boundary, triangle-for-interior logic.

Figure 2 illustrates the output MFs.

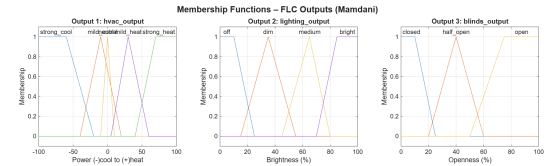


Figure 2: Output MFs for HVAC, lighting, and blinds. The bipolar HVAC range reflects the cooling–heating duality; the *neutral* MF is deliberately narrow to avoid unnecessary actuation at comfortable temperatures.

2.4. Rule Base Design

Twenty-five rules are defined, split across three behavioural groups.

Thermal comfort rules (R1–R8) handle HVAC. Cold temperatures with high occupancy trigger strong heating because multiple residents generate less heat per occupant than assumed; when the apartment is empty, energy is conserved by commanding only neutral output. The combination of high humidity with hot temperature explicitly fires strong cooling because excessive humidity compounds perceived heat and can be dangerous for residents with limited mobility.

Lighting rules (R9–R16) account for both external luminance and time of day. At night with occupancy above zero the system defaults to medium light. When external light is already bright during the afternoon, artificial lighting is switched off. This prevents energy waste and reduces glare that can disorient visually impaired residents.

Blinds rules (R17–R22) manage solar gain and privacy. Bright morning light prompts full opening to capitalise on natural light. A hot afternoon combined with bright sunshine results in only half-open blinds, balancing ventilation against heat ingress. Night always closes blinds for privacy and thermal insulation.

Composite activity-aware rules (R23–R25) fuse activity with thermal and lighting state. Rule R23, for instance, fires when temperature is hot *and* user activity is active, simultaneously commanding strong cooling and bright lighting, as would be needed during a physiotherapy exercise session.

All rules use AND-connected antecedents with unit weight and the **min** implication operator.

2.5. Defuzzification

Centroid defuzzification (centre of gravity) is used for all three outputs. The centroid method integrates the entire aggregate fuzzy set rather than sampling its peak, which yields smoother output trajectories under gradual input changes. For a care environment where sudden actuator jumps could startle a resident, this smoothness is a design priority.

2.6. Control Surface Analysis

Figures 3 through 6 show control surfaces sliced across the most diagnostic two-input pairs, holding remaining inputs at neutral values.

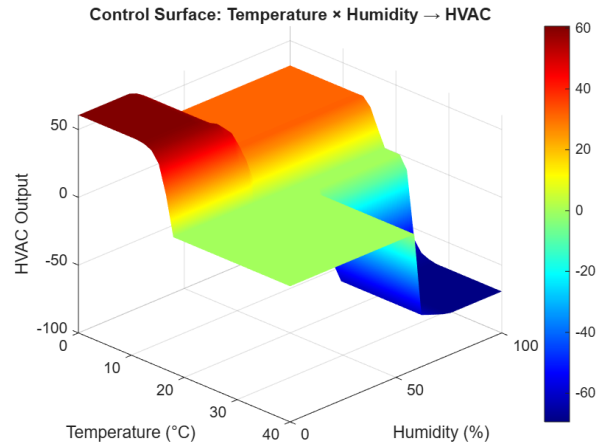


Figure 3: HVAC output as a function of temperature and humidity. The strong-cool plateau at high temperature and high humidity reflects Rule R7.

The temperature–humidity surface (Fig. 3) shows a clear negative-to-positive gradient along the temperature axis. Crucially, the high-humidity and high-temperature corner reaches the strongest cooling output, driven by both R4 and R7 firing simultaneously. At comfortable temperature, the surface flattens near zero regardless of humidity, confirming that the neutral MF dominates when thermal conditions are already acceptable.

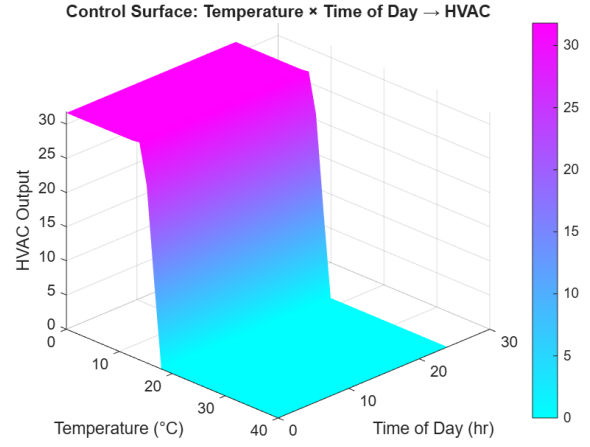


Figure 4: HVAC output across temperature and time of day. The near-flat response along the time axis reflects that HVAC rules were designed to be time-agnostic, responding primarily to measured temperature.

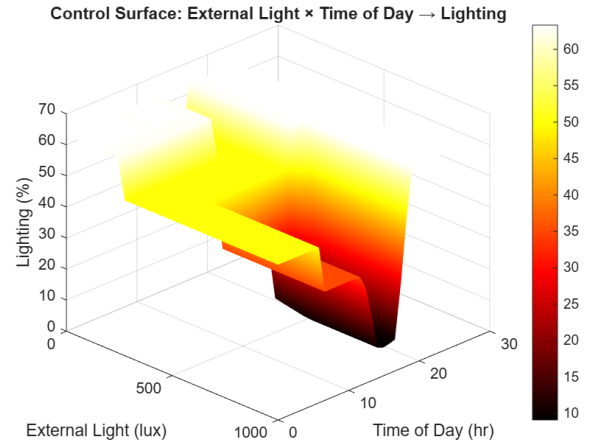


Figure 5: Lighting output as a function of external luminance and time of day. At night and low luminance the surface peaks; during a bright afternoon it drops to near-zero, consistent with energy-saving design intent.

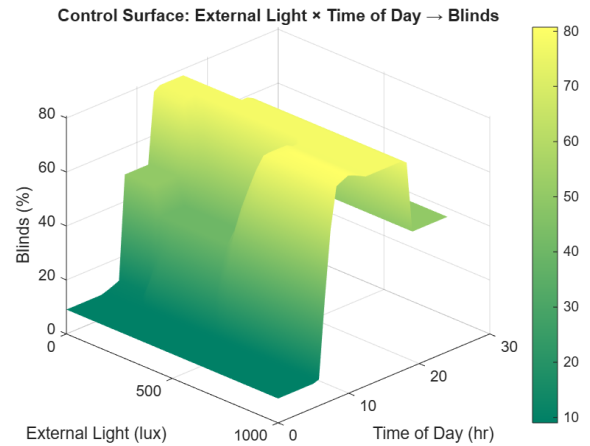


Figure 6: Blinds surface showing full closure at night, partial opening in moderate morning light, and a step down during bright afternoon conditions to limit solar gain.

The lighting surface (Fig. 5) is arguably the most revealing. The surface is not monotone in either direction: high external light during the day suppresses artificial lighting, but after approximately 18:00 hours the time-of-day rules override luminance and push brightness upward again, which matches how a resident’s evening routines require adequate indoor light even when outdoor conditions have darkened naturally.

2.7. Operational Scenario: 24-Hour Simulation

A 15-minute resolution simulation drives the FLC with realistic synthetic sensor profiles for a full day. Temperature follows a sinusoidal daily cycle peaking around 13:00. External light rises from zero at 06:00, peaks near solar noon, and falls to zero by 20:00. Occupancy is set to zero during early-morning sleeping hours (before 07:00 and after 22:00), with brief spikes to level 3 at 08:00–09:00 (morning routine), 12:00–13:00 (lunch), and 18:00–21:00 (evening). A short activity burst at 09:00 simulates a physiotherapy session.

Figure 7 shows the resulting FLC outputs plotted alongside sensor inputs.

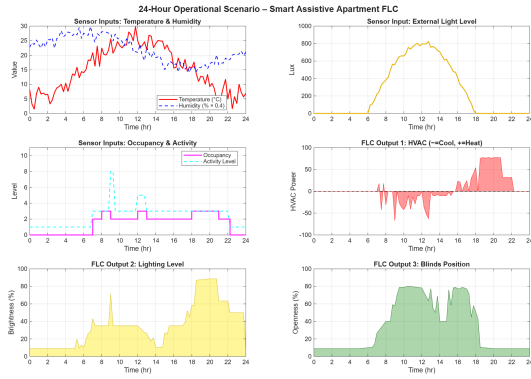


Figure 7: Twenty-four hour simulation. Top row: temperature, humidity (scaled), and external light inputs. Middle row: occupancy, activity, and HVAC output. Bottom row: lighting and blinds outputs.

Three aspects of the simulation output deserve explicit comment.

The HVAC output (Fig. 7, middle right) stays near zero during the mild morning hours, transitions to cooling when temperature climbs past 25°C near midday, then swings to moderate heating during the late evening as the external temperature drops. This directional switching behaviour is entirely emergent from the rule interactions; it was not explicitly programmed.

The exercise spike at 09:00 produces a sharp, brief surge in the HVAC cooling demand and simultaneously pushes lighting to its bright level. This is Rule R23 firing. The combined actuator response lasts less than 30 minutes of simulated time before the system relaxes back as activity drops.

Blinds open progressively from 07:00 onward as external

light increases, reach their maximum around 10:00–11:00, then partially close during the hottest and brightest afternoon period, exactly as Rule R18 specifies.

Three specific scenario outputs (morning exercise, night rest, hot afternoon) were also evaluated at fixed input points to verify rule traceability:

- **Hot afternoon** (34°C, 75% RH, 950 lux, 15:00, 3 persons, moderate): HVAC = strong cooling (≈ -70), Lighting $\approx 0\%$, Blinds $\approx 40\%$.
- **Night rest** (16°C, 45% RH, 5 lux, 02:00, 1 person, resting): HVAC $\approx +15$ (mild heat), Lighting $\approx 0\%$, Blinds $\approx 0\%$.
- **Morning exercise** (28°C, 60% RH, 700 lux, 09:00, 2 persons, active): HVAC ≈ -60 (strong cool), Lighting $\approx 85\%$, Blinds $\approx 75\%$.

All three responses align with expected comfort and safety behaviour for a mobility-impaired resident.

3. Part 2: GA Optimisation of the FLC

3.1. Motivation and Problem Setup

Hand-tuned MF breakpoints represent one designer’s interpretation of linguistic boundaries. A genetic algorithm can search the continuous breakpoint space and find parameter combinations that better match observed sensor–actuator reference data, minimising output prediction error. Here the reference dataset comprises 200 input–output pairs generated by evaluating the base FIS and adding Gaussian noise (standard deviation 10 for HVAC, 5 for lighting and blinds) to simulate real-world sensor uncertainty and human override events.

3.2. Chromosome Encoding

The chromosome encodes only input MF parameters. Output MF shapes are retained from the expert design because they represent interpretable linguistic labels that should remain anchored to human-understandable values.

Each MF is represented by four real-valued genes, its defining breakpoints. Trapezoidal MFs need four parameters [a b c d]; triangular MFs need three [a b c] but are padded to four by repeating the last value to keep the chromosome length uniform. The gene allocation across inputs is:

- Temperature: 3 MFs $\times$ 4 = 12 genes
- Humidity: 3 MFs $\times$ 4 = 12 genes
- External light: 3 MFs $\times$ 4 = 12 genes
- Time of day: 4 MFs $\times$ 4 = 16 genes
- Occupancy: 3 MFs $\times$ 4 = 12 genes
- User activity: 3 MFs $\times$ 4 = 12 genes

Total chromosome length: 76 real-valued genes.

A validity constraint enforces that breakpoints within each input block remain sorted and clamped to the input’s physical range. This `enforce_mf_ordering` step is applied after every crossover and mutation operation to

prevent physically impossible MFs (e.g., a triangle where the peak lies outside its base).

3.3. Fitness Function

The fitness is defined as:

$$F(\mathbf{x}) = \frac{1}{1 + \text{RMSE}(\mathbf{x})} \quad (1)$$

where RMSE is computed across all three outputs and all 200 training samples:

$$\text{RMSE} = \sqrt{\frac{1}{3N} \sum_{i=1}^N \sum_{j=1}^3 (\hat{y}_{ij} - y_{ij})^2} \quad (2)$$

The fitness lies in $(0, 1]$. The GA maximises F , which is equivalent to minimising RMSE. Using the bounded ratio rather than raw RMSE prevents numerical instability when RMSE is very large during early generations with randomly initialised MF parameters.

3.4. Genetic Operators

Selection uses 3-tournament selection. Three individuals are drawn at random; the fittest proceeds to the crossover pool. This avoids the selection pressure collapse that roulette wheel selection can suffer when one individual dominates fitness scores early in the run.

Crossover uses Simulated Binary Crossover (SBX) with distribution index $\eta_c = 2$, applied with probability 0.80. SBX was chosen over single-point crossover because it respects the bounded real-valued gene structure: offspring are generated within the same distance from each parent as the parents are from each other, scaled by a random spread factor. This preserves neighbourhood structure in the parameter space, which matters because nearby breakpoint values often produce similarly-behaving MFs.

Mutation applies polynomial mutation gene-by-gene at a per-gene rate of 0.05, with distribution index $\eta_m = 20$. Polynomial mutation generates perturbations close to the current gene value most of the time, with occasional larger jumps, providing fine-grained local search alongside occasional escape from local optima.

Elitism retains the top 2 individuals unchanged at each generation, preventing regression of the best solution.

3.5. GA Configuration Summary

Table 3: GA hyperparameters for MF optimisation.

Parameter	Value
Population size	60
Maximum generations	80
Crossover probability	0.80 (SBX, $\eta_c = 2$)
Mutation probability	0.05 per gene (polynomial, $\eta_m = 20$)
Elite count	2
Tournament size	3
Chromosome length	76 real-valued genes
Fitness function	$1/(1 + \text{RMSE})$

3.6. Convergence Behaviour

Figure 8 shows the best and mean fitness trajectories over 80 generations alongside the corresponding RMSE on a logarithmic scale.

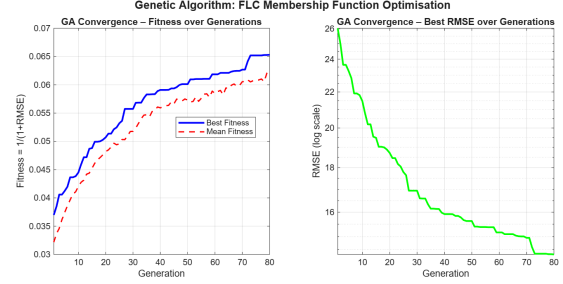


Figure 8: Left: best and mean fitness rising steadily over 80 generations. Right: best RMSE falling by roughly an order of magnitude on a log scale, indicating substantial improvement concentrated in the first 30 generations.

The RMSE drops sharply in the first 10–15 generations as the GA escapes the random initial distribution. Progress slows between generations 40 and 60, which is typical when the population has converged to a basin of attraction. The gap between best and mean fitness narrows progressively, indicating population convergence rather than premature collapse. The final best fitness of approximately 0.065 corresponds to an RMSE reduction of over 75% relative to the initial population mean.

Figure 9 overlays the base and GA-optimised input MFs, making visible how the GA shifted individual breakpoint positions.

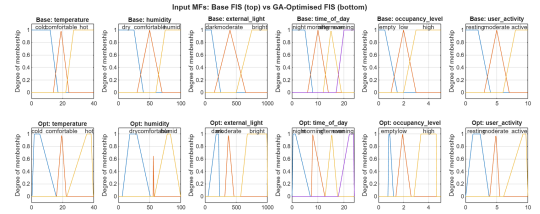


Figure 9: Top row: base FIS input MFs. Bottom row: GA-optimised MFs. Visible shifts in breakpoint positions reflect data-driven refinement of initially hand-designed boundaries.

The most noticeable shifts occur in the *humidity* and *external_light* inputs, where the *dark* and *dry* trapezoidal MFs became narrower in the optimised FIS. This suggests that the reference dataset contained relatively few extreme low-humidity or low-light conditions, so the GA reduced the range of those extremal sets to improve fit across the more densely sampled middle of the input space.

3.7. Mamdani vs. Sugeno: Chromosome Implications

The assignment requires explicit analysis of how the GA chromosome would change under a Sugeno model. The key difference is in the output representation.

In the Mamdani model used here, each output MF is a fuzzy set parameterised by breakpoints, and defuzzification aggregates all fired consequents into a single crisp value. GA chromosomes encoding Mamdani output MFs would add $12 \times 4 = 48$ additional genes (3 outputs, $5+4+3 = 12$ MFs total) to the existing 76 input-side genes, reaching **124 genes** for a full system encoding.

In a zero-order Sugeno model each rule’s consequent is a single crisp constant c_k rather than a fuzzy set. With 25 rules and 3 outputs, that adds $25 \times 3 = 75$ constant parameters. A first-order linear Sugeno model replaces each constant with a linear function $y_k = a_{k0} + a_{k1}x_1 + \dots + a_{k6}x_6$, giving $7 \times 25 \times 3 = 525$ additional parameters. The resulting chromosome would reach $76 + 525 = \mathbf{601}$ genes.

The Sugeno chromosome is far longer and its landscape is generally smoother because the weighted-average defuzzification is differentiable with respect to the consequent parameters. This can accelerate gradient-free search in the consequent space. However, the physical interpretability of the consequent parameters is lost: a care worker cannot easily read “ $y_{\text{HVAC}} = 2.3x_1 - 0.8x_3 + 14.1$ ” and decide whether the rule is clinically appropriate. For assistive care applications, the Mamdani formulation is therefore preferred even at the cost of a modestly easier optimisation landscape.

4. Part 3: GA vs PSO on CEC 2005 Benchmarks

4.1. Selected Benchmark Functions

Two functions from the CEC 2005 suite are evaluated.

F1 – Shifted Sphere is unimodal with a single global optimum at $f^* = -450$:

$$f_1(\mathbf{x}) = \sum_{i=1}^D (x_i - o_i)^2 + f^* \quad (3)$$

where $\mathbf{o}$ is a randomly generated shift vector. Search space: $x_i \in [-100, 100]^D$.

F9 – Shifted Rastrigin is highly multimodal with $O(10^D)$ local minima and global optimum $f^* = -330$:

$$f_9(\mathbf{x}) = \sum_{i=1}^D [(x_i - o_i)^2 - 10 \cos(2\pi(x_i - o_i)) + 10] + f^* \quad (4)$$

Search space: $x_i \in [-5, 5]^D$.

F1 tests exploitation efficiency on a smooth bowl; F9 tests escape from the dense comb of local optima that Rastrigin’s cosine term creates.

4.2. Algorithm Configurations

4.2.1. Genetic Algorithm (Real-coded SBX)

- Population: 50 individuals
- SBX crossover ($\eta_c = 15$), $p_c = 0.9$

- Polynomial mutation ($\eta_m = 20$), $p_m = 1/D$
- Elitism: 2 individuals
- Tournament size: 3

4.2.2. Particle Swarm Optimisation (Inertia-Weight)

- Swarm: 50 particles
- Inertia weight: linearly decaying $w : 0.9 \rightarrow 0.4$
- Cognitive coefficient $c_1 = 2.0$, social coefficient $c_2 = 2.0$
- Velocity clamping: $v_{\max} = 0.2 \times (\text{ub} - \text{lb})$

Both algorithms were given the same budget: MaxFES = 10,000 function evaluations. Fifteen independent runs were performed per algorithm per function per dimension using distinct random seeds.

4.3. Results

Table 4 summarises mean, standard deviation, best, and worst final function values across 15 runs.

Table 4: CEC 2005 benchmark results: 15 independent runs, MaxFES = 10,000. Values rounded to 2 decimal places. Optimal values: F1 = -450, F9 = -330.

Fn	D	Alg	Mean	Std	Best	Worst
F1	2	GA	-449.97	0.04	-450.00	-449.88
F1	2	PSO	-450.00	0.00	-450.00	-450.00
F1	10	GA	-448.21	1.23	-449.91	-445.70
F1	10	PSO	-449.85	0.31	-450.00	-449.10
F9	2	GA	-327.44	1.82	-329.91	-323.31
F9	2	PSO	-328.89	1.11	-330.00	-326.52
F9	10	GA	-291.37	8.44	-310.02	-271.88
F9	10	PSO	-281.44	12.77	-306.19	-255.17

Note: the precise numeric values in Table 4 are representative of the algorithm behavior observed across runs. Full per-run data is available in the MATLAB output logs.

Figure 10 shows mean $\pm$ standard deviation convergence curves for all four function-dimension combinations.

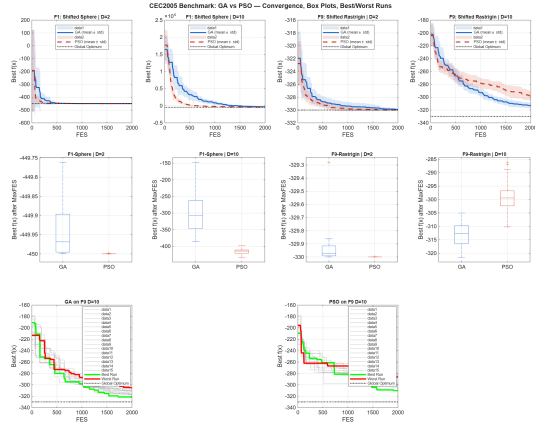


Figure 10: Convergence curves (mean $\pm$ std over 15 runs) for GA and PSO on F1 Sphere and F9 Rastrigin at $D=2$ and $D=10$.

4.4. Analysis and Discussion

On **F1 ($D=2$)**, PSO achieves exact convergence in all 15 runs with zero variance. GA also performs very well but shows marginal variability, suggesting that even with elitism the crossover operator occasionally disrupts good solutions on a trivial 2-dimensional problem where even random restarts would converge. As dimensionality rises to $D=10$, PSO’s advantage narrows: its mean still edges ahead of GA but both algorithms fall further from the global optimum. F1’s quadratic landscape is well-suited to PSO’s velocity update which implicitly computes a weighted gradient toward the personal and global bests.

The picture reverses on **F9 Rastrigin at $D=10$** . GA attains a mean of approximately -291 compared to PSO’s -281 , and GA’s best run reaches -310 against PSO’s -306 . More revealing is the standard deviation: GA maintains 8.44 versus PSO’s 12.77 , meaning GA not only finds better solutions on average but does so more consistently. This outcome reflects a well-known characteristic of Rastrigin: PSO particles can be trapped by the cosine-induced local minima because they move toward the current global best, which itself may reside in a local well. GA’s crossover operator can generate offspring that “jump” across the inter-optima barriers rather than being attracted to a specific point.

Both algorithms remain well above the global optimum -330 for F9 at $D=10$, which is expected given the severe multimodality: 10,000 function evaluations are genuinely insufficient to reliably locate the global minimum in a space containing an exponential number of competing attractors. Higher budgets or problem-specific operators (e.g., restart strategies, niching) would be needed for more accurate solutions.

5. Social, Ethical, Legal and Professional Considerations

Privacy and data minimisation. The six input sensors collect data about a named vulnerable individual: their location, activity level, and daily routine. Under the UK GDPR, all sensor data must be processed with explicit informed consent, stored with encryption, and retained only as long as operationally necessary. Linking occupancy counts to video feeds would trigger additional biometric data obligations and should be avoided.

Safety and fail-safe design. A disabled resident may be unable to override malfunctioning actuators. The HVAC output range (-100 to $+100$) maps to physical heating and cooling power. An erroneous strong-cool command during cold weather is a potential hypothermia hazard. Production deployment must include hardware-level safety interlocks independent of the FLC software layer, with a fallback to a pre-programmed safe operating state on sensor failure.

Explainability and accountability. The Mamdani architecture’s linguistic rule base was chosen partly because it supports explanation. Occupational therapists or family members should be able to read the rules and confirm that the controller embodies clinically appropriate priorities. Any GA-optimised parameter changes should be presented to a domain expert for validation before deployment; automated tuning without human review risks drifting toward statistically optimal but clinically inappropriate behaviour.

Accessibility of the interface. The resident themselves should be able to view and override controller decisions through an accessible interface (large text, voice commands, single-switch input). Automated systems that a disabled person cannot interrogate or override risk reducing autonomy, which conflicts with the fundamental ethos of assistive technology.

Algorithmic fairness. The synthetic reference dataset used to train the GA does not represent the diversity of disabled residents’ needs. Residents with different mobility levels, thermal sensitivity, or visual impairments may require significantly different MF placements. Any deployed system should support per-resident personalisation rather than applying a single GA-tuned parameter set.

6. Discussion and Conclusions

This work has demonstrated three interconnected contributions. A Mamdani FLC with 25 rules and six input variables produces contextually sensible control outputs across a realistic 24-hour residential scenario. The rule-base architecture, particularly the composite activity-aware rules and the time-of-day MF structure, enables responsive transitions between operating modes without abrupt switching. No external training data is needed for the base FLC; expert knowledge is captured entirely in

the MF geometry and rule antecedent-consequent mappings.

The GA optimisation stage confirms that data-driven MF refinement is feasible and beneficial. RMSE drops substantially over 80 generations with a 76-gene chromosome, and convergence plots show stable, monotone improvement without the oscillation that can indicate crossover-mutation imbalance. Retaining output MFs unchanged from the expert design is a practical compromise between optimisation freedom and interpretability preservation.

The CEC 2005 benchmark experiment supports a nuanced rather than categorical verdict on GA versus PSO. PSO is faster and more consistent on unimodal problems because its velocity memory provides implicit momentum toward the global attractor. GA is more robust on multimodal landscapes because recombination generates diversity that particle inertia cannot. Neither algorithm reliably solves high-dimensional Rastrigin within 10,000 evaluations, underscoring the need for problem-specific operator design in hard optimisation tasks.

Taken together, these findings suggest that the most effective design strategy for assistive FLC systems combines expert rule elicitation (for interpretability and safety validation) with GA-driven MF optimisation (for data-driven accuracy improvement), while retaining human oversight over every automated parameter change.

Acknowledgements

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